

LOVELY PROFESSIONAL UNIVERSITY

Sample Question Set - Unit 1

Course Code: CSE276

Course Title: ARTIFICIAL INTELLIGENCE
FOUNDATIONS

Course Outcome: CO1

Paper: Sample Subjective Questions

Maximum Marks: 10 x 10 = 100

Unit: 1

Unit Title: Foundations of AI and AI Problem Solving

Unit Mapped RBT: L2 (Understand)

Total Questions: 10

Suggested Duration: Not timed - practice set

Subjective Questions (10 x 10 = 100 Marks)

Q1. Explain the relationship between Artificial Intelligence, Machine Learning, Deep Learning and Data Science. (10 Marks)

- (a) Describe the scope of each of the four and state how they are related to one another.
- (b) Explain, with one example, how a system can belong to Artificial Intelligence without involving any machine learning.

[L2 - Understand]

Q2. Describe the classification of Artificial Intelligence by capability and by functionality. (10 Marks)

- (a) Name and describe the classes that make up each of the two axes.
- (b) Explain which of those classes are represented by systems in service today and which are not.

[L2 - Understand]

Q3. Distinguish the search space of a problem from its solution space. (10 Marks)

- (a) Describe what each of the two contains.
- (b) Explain, using a route-planning task, why the search space is normally very much the larger of the two.

[L2 - Understand]

Q4. Explain what the characteristics of a problem tell a designer before any technique is chosen. (10 Marks)

- (a) Describe three problem characteristics and what each one asks about the problem.
- (b) Explain how each of the three narrows the choice of method, using a named task as the example.

[L2 - Understand]

Q5. Describe the applications of Artificial Intelligence in healthcare and in agriculture. (10 Marks)

- (a) Describe two applications drawn from each of the two sectors.
- (b) Explain what the input data looks like in one of them, and one risk that the application carries.

[L2 - Understand]

Q6. Explain the performance measures used to compare two search algorithms. (10 Marks)

- (a) Describe completeness, optimality, time complexity and space complexity.
- (b) Explain how an algorithm can be complete without being optimal, giving an example.

[L2 - Understand]

Q7. Summarize the history of Artificial Intelligence through the two winters it has passed through. (10 Marks)

- (a) Describe what brought about each of the two winters.
- (b) Explain what each of the two revealed about the approach that was in favour at the time.

[L2 - Understand]

Q8. Interpret the following task using the tools of problem formulation. A delivery company wishes to plan a round that visits twelve addresses and returns to the depot. (10 Marks)

- (a) Describe how the task is written as a state-space problem, naming each component.
- (b) Explain why the size of the state space makes an exhaustive search impractical here.

[L2 - Understand]

Q9. Explain why an Artificial Intelligence system is monitored after release rather than treated as finished. (10 Marks)

- (a) Describe two things that can change after a system has been deployed.
- (b) Explain what should be measured, and what a poor reading would prompt the team to do.

[L2 - Understand]

Q10. Describe how Artificial Intelligence is applied in smart cities and in education. (10 Marks)

- (a) Describe two applications drawn from each of the two areas.
- (b) Explain one ethical concern that arises in one of the applications you have described.

[L2 - Understand]

----- End of Sample Question Set -----